

Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

1 Marginal PMF

1.1 Definition

For two discrete random variables X and Y , the joint PMF is defined as:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

Key Idea (from lecture)

The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

1.2 Step-by-Step Derivation

To extract marginal distributions:

For X :

$$p_X(x) = \sum_y p_{XY}(x, y)$$

Reasoning:

- Fix x
- Sum probabilities over all possible y
- This accounts for all outcomes where $X = x$

For Y :

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Reasoning:

- Fix y
- Sum over all x
- This captures all events where $Y = y$

1.3 Marginal PMF from Joint PMF Table

From the table (page 4):

- Row sums $\rightarrow p_X(x)$
- Column sums $\rightarrow p_Y(y)$

Mathematically:

$$p_X(x_i) = \sum_j p_{ij}, \quad p_Y(y_j) = \sum_i p_{ij}$$

Normalization Property:

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

1.4 Worked Example (Step-by-Step)

From the table (page 5):

$X \setminus Y$	$y = 0$	$y = 1$
$x = 0$	1/8	1/4
$x = 1$	3/8	1/4

Step 1: Compute $p_X(x)$

For $x = 0$:

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{1}{8} + \frac{2}{8} = \frac{3}{8}$$

For $x = 1$:

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$

Step 2: Compute $p_Y(y)$

For $y = 0$:

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$$

For $y = 1$:

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Final Marginals

$$p_X(x) = \begin{cases} 3/8, & x = 0 \\ 5/8, & x = 1 \end{cases} \quad p_Y(y) = \begin{cases} 1/2, & y = 0 \\ 1/2, & y = 1 \end{cases}$$

2 Correlation

2.1 Definition

Correlation is introduced via the product moment:

$$R_{XY} = E[XY]$$

2.2 Interpretation (Step-by-Step)

From lecture (pages 6–7):

- Measures average value of product XY
- Depends on:
 - Mean values
 - Spread
 - Dependence

Important Observation

$$R_{XY} = 0 \Rightarrow \text{orthogonal about origin}$$

2.3 Limitation

- Computed about the origin (not means)
- Mixes: mean, variance, dependence

$\Rightarrow$ Therefore, it is not a clean measure of dependence.

Transition to Covariance: To remove mean effects:

$$\text{Center variables} \Rightarrow \text{Covariance}$$

3 Covariance

3.1 Definition

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Step-by-Step Meaning

- **Subtract means:** $(X - \mu_X)$ and $(Y - \mu_Y)$
- **Multiply deviations:** Measures joint variation
- **Take expectation:** Average behavior

3.2 Interpretation

From page 9:

- $\text{Cov}(X, Y) > 0$: same direction
- $\text{Cov}(X, Y) < 0$: opposite direction
- $\text{Cov}(X, Y) = 0$: no linear relation

Warning: Magnitude depends on units $\rightarrow$ cannot measure strength directly.

3.3 Algebraic Derivation (Step-by-Step)

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y] \quad (\text{Expand product}) \\ &= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y \quad (\text{Apply expectation linearly}) \\ &= E[XY] - \mu_X\mu_Y - \mu_X\mu_Y + \mu_X\mu_Y \quad (\text{Substitute } E[X] = \mu_X, E[Y] = \mu_Y) \\ &= E[XY] - \mu_X\mu_Y\end{aligned}$$

Final Formula

$$\boxed{\text{Cov}(X, Y) = E[XY] - E[X]E[Y]}$$

3.4 Properties

- **Symmetry:** $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- **Variance case:** $\text{Cov}(X, X) = \text{Var}(X)$
- **Linearity:** $\text{Cov}(aX + b, cY + d) = ac \cdot \text{Cov}(X, Y)$
- **Independence:** $X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$

Warning: Zero covariance $\neq$ independence (in the general case).

3.5 Example 1

Given: $Y = -2X + 3$

Step 1: Formula

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 2: Compute XY

$$XY = X(-2X + 3) = -2X^2 + 3X$$

Step 3: Compute $E[Y]$

$$E[Y] = E[-2X + 3] = -2E[X] + 3$$

Step 4: Substitute

$$\begin{aligned}\text{Cov}(X, Y) &= E[-2X^2 + 3X] - E[X](-2E[X] + 3) \\ &= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X] \\ &= -2E[X^2] + 2(E[X])^2\end{aligned}$$

Step 5: Use variance

$$\begin{aligned}\text{Var}(X) &= E[X^2] - (E[X])^2 \\ \Rightarrow \text{Cov}(X, Y) &= -2\text{Var}(X)\end{aligned}$$

4 Pearson's Correlation Coefficient

4.1 Definition

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

4.2 Interpretation

- **Measures:** Direction and Strength
- **Properties:** Unit-free, Normalized covariance
- **Range:** $-1 \leq \rho_{XY} \leq 1$

4.3 Meaning

- $\rho > 0$: positive linear trend
- $\rho < 0$: negative linear trend
- $\rho = 0$: no linear correlation

5 Joint Gaussian Random Variables

5.1 Definition

A pair (X, Y) is jointly Gaussian if:

- Joint PDF has bivariate Gaussian form
- Marginals are Gaussian

5.2 Algebraic Form

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho \left(\frac{x-\mu_X}{\sigma_X}\right) \left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]\right)$$

5.3 Parameters

- μ_X, μ_Y : means
- σ_X, σ_Y : standard deviations
- ρ : correlation coefficient

5.4 Interpretation (from lecture example)

Example (page 23):

- X : CO₂ concentration
- Y : temperature

If $\rho > 0 \Rightarrow$ Higher CO₂ $\rightarrow$ higher temperature.

5.5 Applications

- Signal processing
- Communication systems
- Modeling noisy measurements

Final Concept Flow (Important for Exam)

Joint PMF $\rightarrow$ Marginal PMF (via summation) $\rightarrow$ Product moment ($E[XY]$) $\rightarrow$ raw correlation $\rightarrow$ Centering $\rightarrow$ Covariance $\rightarrow$ Normalize $\rightarrow$ Pearson coefficient $\rightarrow$ Continuous extension $\rightarrow$ Joint Gaussian